

1.Functional Requirements

Functional requirements describe the main features of the application and what the system should do.

Admin

- * Login securely to the system.
- * Add, edit, and delete bus details.
- * Manage driver information.
- * Manage student and parent details.
- * Assign buses and routes to students.
- * Monitor the live location of all buses.
- * View trip history and reports.

Driver

- * Login securely to the application.
- * Start and end the bus trip.
- * Share the live GPS location of the bus.
- * Update the bus status (On Time, Delayed, Arrived).
- * Send an emergency alert if needed.

Parent

- * Login to the application.
- * View the live location of the school bus.
- * Receive notifications for bus departure, arrival, delays, and trip completion.
- * Check the estimated arrival time (ETA).

* View previous trip details.

2.Non-Functional Requirements

Non-functional requirements describe how well the system should perform.

- * Easy to Use: The application should have a simple and user-friendly interface.
- * Fast Performance: Bus location and notifications should be updated quickly.
- * Secure: Only authorized users should be able to access the system.
- * Reliable: The system should provide accurate GPS tracking and notifications.
- * Available: The application should be accessible whenever users need it.
- * Scalable: The system should support multiple buses, drivers, students, and schools.
- * Maintainable: The application should be easy to update and improve in the future.

3.Technology Stack

Technology	Purpose
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Java	Application development
Android Studio	Mobile app development
XML	User interface design
Firebase / MySQL	Store application data
Google Maps API	Display live bus location
Firebase Cloud Messaging (FCM)	Send notifications to parents
Firebase Authentication	Secure user login
Git & GitHub	Version control and project management

4. Project Modules

1. Admin Module

The Admin manages the entire transportation system.

- * Login
- * Manage buses
- * Manage drivers
- * Manage students
- * Assign routes
- * Monitor live bus locations
- * Generate reports

2. Driver Module

The Driver operates the bus and shares its live location.

- * Login
- * Start the trip
- * Share live GPS location
- * Update trip status
- * End the trip
- * Send emergency alerts

3. Parent Module

The Parent tracks the bus and receives updates.

- * Login
- * View live bus location
- * Receive notifications
- * Check estimated arrival time (ETA)
- * View trip history

4. GPS Tracking Module

This module tracks the bus in real time.

- * Collect live GPS location
- * Update bus position continuously
- * Display the bus on Google Maps
- * Track the bus route

5. Notification Module

This module keeps parents informed.

- * Bus departure notification
- * Bus arrival notification
- * Delay notification
- * Trip completion notification
- * Emergency alerts

6. Database Module

This module stores all application data.

- * Student information

- * Parent information
- * Driver details
- * Bus details
- * Route details
- * Trip history

5.Module Overview

Module	Description
Admin Module	Manages buses, drivers, students, and routes.
Driver Module	Updates the bus location and trip status.
Parent Module	Allows parents to track the bus and receive notifications.
GPS Tracking Module	Tracks the live location of the school bus.
Notification Module	Sends alerts about bus movements and delays.
Database Module	Stores and manages all transportation data.